ClipSAT : Math for AI & Data Science

Unit 1.01: Vectors & Embeddings in AI

Name: _____

Q1. Compute the dot product of the embedding vector mathbf{u} = [2, 3]^T with itself.

[ ] A: 13
[ ] B: 15
[ ] C: 11
[ ] D: 10

Q2. Compute the dot product of the embedding vector mathbf{u} = [3, 4]^T with itself.

[ ] A: 25
[ ] B: 29
[ ] C: 23
[ ] D: 14

Q3. Compute the dot product of the embedding vector mathbf{u} = [4, 5]^T with itself.

[ ] A: 41
[ ] B: 47
[ ] C: 39
[ ] D: 18

Q4. Compute the dot product of the embedding vector mathbf{u} = [5, 6]^T with itself.

[ ] A: 61
[ ] B: 69
[ ] C: 59
[ ] D: 22

Q5. Compute the dot product of the embedding vector mathbf{u} = [6, 7]^T with itself.

[ ] A: 85
[ ] B: 95
[ ] C: 83
[ ] D: 26

Q6. Compute the dot product of the embedding vector mathbf{u} = [7, 8]^T with itself.

[ ] A: 111
[ ] B: 125
[ ] C: 113
[ ] D: 30

Q7. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [2, -3]^T .

[ ] A: |mathbf{w}|_1 = 5, ; |mathbf{w}|_2 =
[ ] B: |mathbf{w}|_1 = -1, ; |mathbf{w}|_2
[ ] C: |mathbf{w}|_1 = 5, ; |mathbf{w}|_2 =
[ ] D: |mathbf{w}|_1 = sqrt{13}, ; |mathbf{

Q8. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [4, -6]^T .

[ ] A: |mathbf{w}|_1 = -2, ; |mathbf{w}|_2
[ ] B: |mathbf{w}|_1 = 10, ; |mathbf{w}|_2
[ ] C: |mathbf{w}|_1 = 10, ; |mathbf{w}|_2
[ ] D: |mathbf{w}|_1 = sqrt{52}, ; |mathbf{

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